Data Analyst Job Postings Analysis

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Introduction

This project analyzes a sample of Data Analyst job postings to explore how salary estimates relate to minimum years of experience and programming language requirements. The analysis uses median salary estimates, extracted minimum years of experience, and data language categories such as R, Python, Both, and Neither.

Years of Experience Analysis

The scatter plot below compares Minimum Years Experience on the x-axis with Median Salary Estimate on the y-axis. Rows with non-numeric or missing experience values were excluded from this plot.

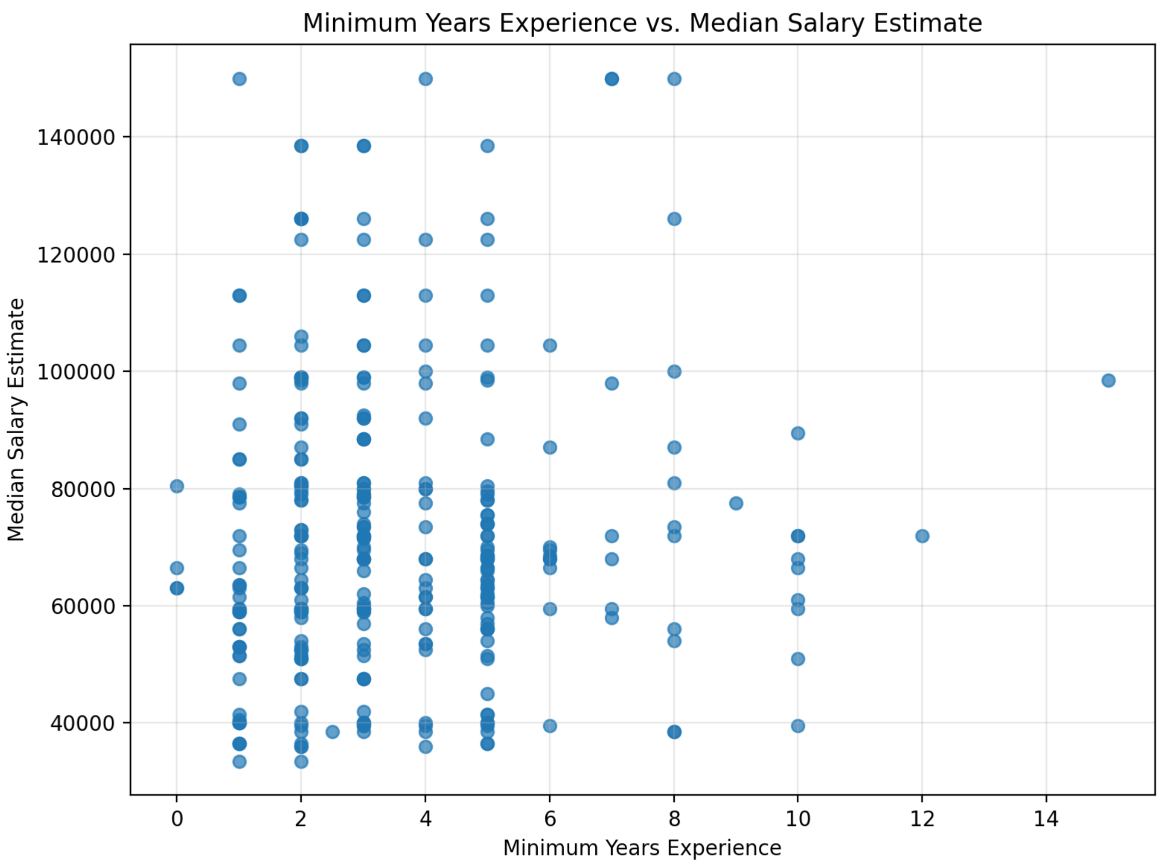

Figure 1. Minimum Years Experience vs. Median Salary Estimate

Interpretation: The scatter plot shows a wide spread of salary estimates across different experience levels. Although some higher-salary jobs require more years of experience, the overall relationship appears weak because many roles with similar experience requirements have very different salary estimates. This suggests that salary may also depend on other factors such as location, company size, industry, job responsibilities, and required technical skills.

Programming Language Analysis

The box plot below compares Median Salary Estimate across the Data Languages groups.

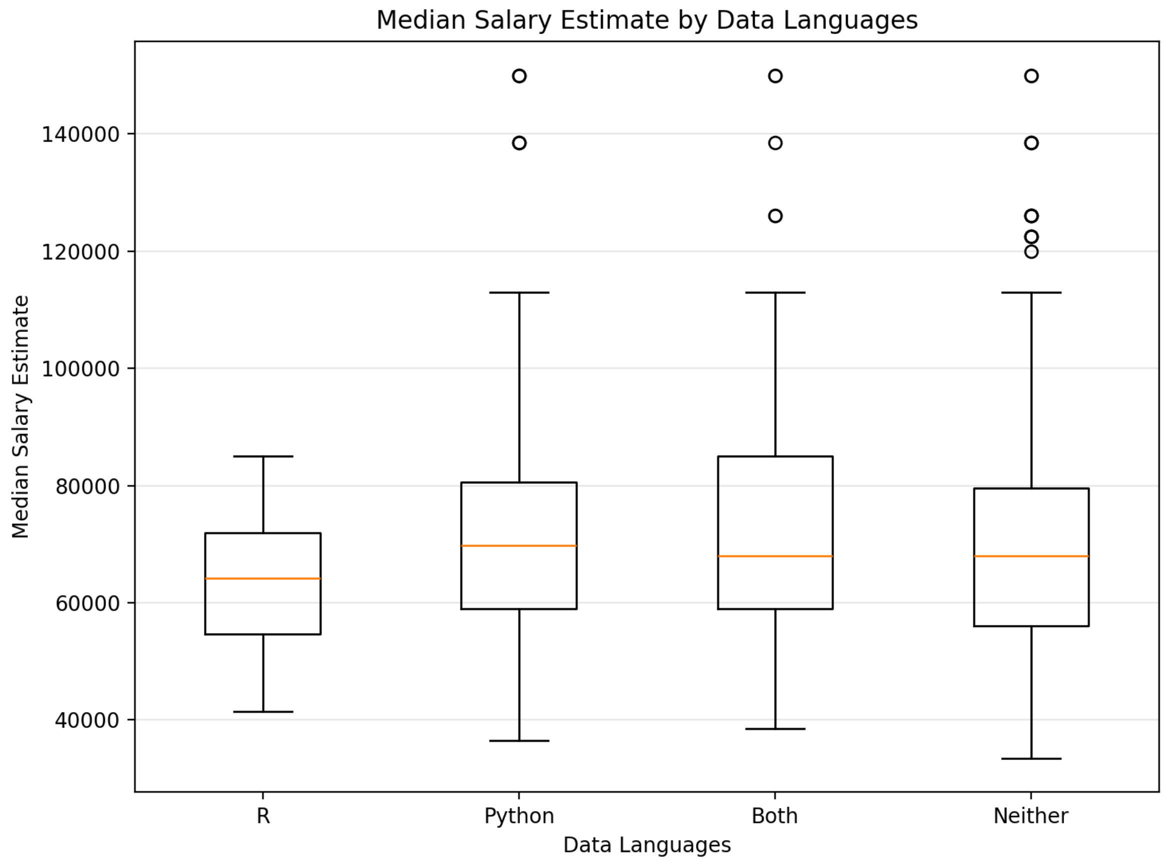

Figure 2. Median Salary Estimate by Data Languages

Interpretation: The box plot shows that postings mentioning Python, Both Python and R, or Neither have fairly similar median salary levels, while the R-only group has fewer postings and a slightly lower median in this sample. Python and Both also show several high-salary outliers, suggesting that roles involving programming skills may sometimes offer higher compensation, although the categories overlap a lot and the sample size for R-only postings is small.

Reflection

I began the workshop 3 assignment with some anxiety, as I found both the pre-workshop materials and the live session quite technical. Unfortunately, I couldn't complete the session due to family obligations. However, I eventually pushed myself to start the assignment.

Once I got going, it turned out to be a smooth and enjoyable task. The instructions were straightforward, but I made a mistake in Part 1B, question 2. I used the same cell (C2) for all five tested rows, which resulted in getting the same outcome (5 years of minimum experience) even when it was incorrect. I realized my error after some thought and was able to get back on track. From that point onward, everything went smoothly.

I continue to be impressed by AI's capacity and its ability to work with large datasets at incredible speed.